Intractability of Approximate Correlated Equilibria in Two-Player Convex Games

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Abstract

We show that computing an ϵ-approximate normal-form correlated equilibrium (ϵ-CE) in two-player convex games admits no fully polynomial-time approximation scheme (FPTAS), resolving a central open question of Anagnostides et al. [1]. Specifically, in the standard first-order black-box oracle model on the unit Euclidean balls $\mathcal{X}_1 = B_{d_1}(\mathbf{0}, 1)$ and $\mathcal{X}_2 = B_{d_2}(\mathbf{0}, 1)$ ($d = d_1 + d_2$), we construct a family of $\frac{1}{2}$-strongly convex quadratic games with costs in $[0, 1]$ and Euclidean Lipschitz constant $L < \sqrt{d}$ on which every randomized algorithm outputting a $\frac{1}{d}$-CE with probability at least $\frac{2}{3}$ requires at least $\exp(\Omega(d^{1/16})) = d^{\omega(1)}$ oracle queries—regardless of internal running time or the support size of the output distribution.

At the core of our construction is a self-map $f_{\mathcal{V}} : \mathbb{R}^{d_0} \to B_{d_0}(\mathbf{0}, \frac{1}{16})$ whose fixed-point iterates trace an exact N-step post-order traversal ($N = 3 \cdot 2^k - 2$) of a recursive binary-unary tree $\mathcal{F}_k$ along near-orthogonal Rademacher vectors. Two combinatorial properties drive the proof: (1) an exact telescoping partition of unity $\sum_{s \in \mathcal{F}_k} L_s(w) = 1 - w_{r_k}$ across all continuous activations $w \in [0, 1]^{\mathcal{F}_k}$, and (2) a post-order stack identity showing that for each vertex $s' \in \mathcal{F}_k$, the active set $B(s')$ partitions the post-order prefix $\{u \leq_{\text{post}} s'\} = \bigsqcup_{b \in B(s')} \text{Subtree}(b)$ into disjoint subtrees, which yields the exact upper-triangular intersection matrix $|\text{Path}(s, r_k) \cap B(s')| = \mathbb{I}[s \leq_{\text{post}} s']$. Contracting the two-player conditional-expectation residual $Af_{\mathcal{V}}(X_{\text{act}}) - X_{\text{act}}$ against the non-conservative test vector field $G(x) = \sum_{s \in \mathcal{F}_k} L_s(w(x))u_s$ (with $u_s = \sum_{a \in \text{Path}(s, r_k)} v_a$) produces a strictly positive symmetrized drift exceeding $\frac{9}{32}\alpha$ across all active pairs in the pre-terminal slab $\{|\langle x, v_{r_k}\rangle| \leq \tau_0\}$, forcing any algorithm that has not yet discovered the hidden root vector v_{r_k} to incur swap regret exceeding $\frac{1}{d}$.

1 Introduction

A two-player convex game $G = (\mathcal{X}_1, \mathcal{X}_2, c_1, c_2)$ consists of compact convex strategy sets $\mathcal{X}_i \subseteq B_{d_i}(\mathbf{0}, R) \subset \mathbb{R}^{d_i}$ ($d = d_1 + d_2$) and L-Lipschitz cost functions $c_i : \mathcal{X}_1 \times \mathcal{X}_2 \to [0, 1]$ that are convex in player i's own strategy x_i for each fixed opponent strategy x_{-i}. A fundamental divide separates tractable and potentially intractable correlated solution concepts in such games:

- On the positive side, *coarse correlated equilibria*, *linear correlated equilibria*, and poly-dimensional Φ-*equilibria* all admit algorithms running in time $\text{poly}(d, R, L, 1/\epsilon)$ [5, 4, 1].
- On the other hand, for full *normal-form correlated equilibria* (ϵ-CE)—where each player demands low swap regret against *all* strategy modifications $\phi_i : \mathcal{X}_i \to \mathcal{X}_i$—all known algorithms require $\exp(\tilde{\Omega}(1/\epsilon))$ oracle queries via the `TreeSwap` framework [7, 3].

Recently, Anagnostides et al. [1] (Theorem 3.1) showed that computing an ϵ-CE in two-player convex quadratic games is as hard as finding an ϵ'-approximate fixed point of an unknown norm

contraction (UnkContr). However, that reduction requires exponentially small precision $\epsilon = \Theta(\gamma^2\delta^2)$ when the contraction modulus γ is close to 1, leaving open whether two-player convex games admit an FPTAS running in time $\mathrm{poly}(d, R, L, 1/\epsilon)$ at inverse-polynomial error $\epsilon = 1/\mathrm{poly}(d)$.

1.1 Main Result

We resolve this question in the negative: even on exact unit Euclidean balls $\mathcal{X}_i = B_{d_i}(\mathbf{0}, 1)$ with $\frac{1}{2}$-strongly convex quadratic costs and sublinear Lipschitz modulus $L < \sqrt{d}$, finding a $\frac{1}{d}$-CE requires superpolynomial ($\exp(\Omega(d^{1/16}))$) oracle calls.

Theorem 1 (Informal Version of Theorem 2)**.** *For every sufficiently large ambient dimension $d = d_1 + d_2$, there exists a finite family of two-player convex games on $\mathcal{X}_1 = B_{d_1}(\mathbf{0}, 1)$ and $\mathcal{X}_2 = B_{d_2}(\mathbf{0}, 1)$ whose costs $c_1, c_2 : \mathcal{X}_1 \times \mathcal{X}_2 \to [0, 1]$ are $\frac{1}{2}$-strongly convex in each player's own strategy and L-Lipschitz with $L < \sqrt{d}$, such that any randomized black-box algorithm outputting a $\frac{1}{d}$-CE with probability at least $\frac{2}{3}$ requires at least $\frac{1}{2}\exp\big(\Omega(d^{1/16})\big) = d^{\omega(1)}$ first-order queries.*

1.2 Proof Overview and Overcoming Three Structural Barriers

Our hard game family implements an asymmetric imitation-chasing dynamic on active coordinates $x_{\mathrm{act}}, y_{\mathrm{act}} \in \mathbb{R}^{d_0}$ padded to unit balls $\mathcal{X}_i = B_{d_i}(\mathbf{0}, 1)$:

$$c_1(x, y) \;=\; \frac{1}{4}\big\|x_{\mathrm{act}} - y_{\mathrm{act}}\big\|_2^2 + \frac{1}{4}\|x_{\mathrm{pad}}\|_2^2, \qquad c_2(x, y) \;=\; \frac{1}{4}\big\|y_{\mathrm{act}} - f_{\mathcal{V}}(x_{\mathrm{act}})\big\|_2^2 + \frac{1}{4}\|y_{\mathrm{pad}}\|_2^2.$$

Here Player 1 wants x_{act} to match Player 2's action y_{act}, while Player 2 wants y_{act} to match the next-step target $f_{\mathcal{V}}(x_{\mathrm{act}})$. On any ϵ-CE μ, Player 1 can deviate to the conditional expectation $PY_{\mathrm{act}}(x_{\mathrm{act}}) = \mathbb{E}_\mu[Y_{\mathrm{act}} \mid X_{\mathrm{act}} = x_{\mathrm{act}}]$ and Player 2 can deviate to $P^\star f_{\mathcal{V}}(X_{\mathrm{act}})(y_{\mathrm{act}}) = \mathbb{E}_\mu[f_{\mathcal{V}}(X_{\mathrm{act}}) \mid Y_{\mathrm{act}} = y_{\mathrm{act}}]$. Composing the two conditional expectation operators yields a self-adjoint contraction $A := PP^\star$ on $L^2(\mu_{1,\mathrm{proj}}; \mathbb{R}^{d_0})$ for which low swap regret implies the approximate fixed-point equation

$$\big\|Af_{\mathcal{V}}(X_{\mathrm{act}}) - X_{\mathrm{act}}\big\|_{L^2(\mu_{1,\mathrm{proj}})} \;\le\; 4\sqrt{\epsilon}.$$

Turning this L^2 residual bound into an unconditional $\exp(\Omega(d^{1/16}))$-query lower bound at inverse-linear precision $\epsilon = \frac{1}{d}$ requires overcoming three classical obstacles simultaneously:

1. **The Halpern Iteration Barrier:** If $f_{\mathcal{V}}$ were nonexpansive in *any* norm $\|\cdot\|_\star$, anchored Halpern iteration $x_{n+1} = \frac{1}{n+2}x_0 + \frac{n+1}{n+2}f_{\mathcal{V}}(x_n)$ would find an approximate fixed point $\|x_n - f_{\mathcal{V}}(x_n)\|_\star \le \sqrt{\epsilon}$—and hence an ϵ-CE—in $O(1/\sqrt{\epsilon}) = O(\sqrt{d})$ queries [6, 2]. Thus $f_{\mathcal{V}}$ must have Lipschitz constant $\mathrm{Lip}(f_{\mathcal{V}}) \gg 1$ (yet still $\mathrm{poly}(d)$ so c_2 remains $O(\sqrt{d})$-Lipschitz), which we achieve using narrow piecewise-linear ramp transitions of width $\delta_w = \frac{\alpha}{8(2k+1)}$ combined with a Cauchy–Schwarz active-sparsity bound showing at most $O(\tau_0^{-2}) = O(k^4)$ tree nodes can activate at any point in $B_{d_0}(\mathbf{0}, 1)$.

2. **The Single-Block Parseval–Brouwer Barrier:** If moving along an N-step walk required keeping N near-orthogonal directions active simultaneously with inner products $\ge \alpha$, Bessel's inequality would force norm $\sqrt{N}\,\alpha \le 1$, shrinking α to $O(N^{-1/2})$ and driving α^2 below $\frac{1}{d}$ whenever $N \gg d$. We circumvent this via space compression on a recursive tree $\mathcal{F}_k$ of depth parameter k and size $N = 3 \cdot 2^k - 2$: at every step t, the active set $B(s_t)$ stores only the roots of maximal completed subtrees—at most $k + 1$ vectors—allowing $\alpha = \frac{1}{16(k+1)}$ to remain inverse-logarithmic in N while all iterates stay inside $B_{d_0}(\mathbf{0}, \frac{1}{16})$.

3. **The Orbit-Averaging / Conservative Potential Barrier:** As observed in [1] (Section 5.5), taking the uniform average over the first $M = \lceil 2/\epsilon \rceil$ pure trajectory steps satisfies $\mathbb{E}[\Phi(f_{\mathcal{V}}(X)) - \Phi(X)] = \frac{\Phi(x^{(M)}) - \Phi(x^{(0)})}{M} \leq \epsilon$ for *every* bounded scalar potential Φ. Hence contracting $Af_{\mathcal{V}}(X_{\mathrm{act}}) - X_{\mathrm{act}}$ against any conservative gradient field $\nabla\Phi(X_{\mathrm{act}})$ can never prove a query bound larger than $O(1/\epsilon) = O(d)$!

   To break this telescoping cancellation, we contract $Af_{\mathcal{V}}(X_{\mathrm{act}}) - X_{\mathrm{act}}$ against a *non-conservative* test vector field

$$G(x) \;=\; \sum_{s \in \mathcal{F}_k} L_s(w(x))\, u_s, \qquad \text{where } u_s \;:=\; \sum_{a \in \mathrm{Path}(s, r_k)} v_a.$$

   Here two structural identities work in tandem:

   - First, on the negative side ($-\langle G(x), x\rangle$), whenever a tree vertex a has nonzero subtree weight $S_a(w(x)) = \sum_{s \in \mathrm{Subtree}(a)} L_s(w(x)) > 0$, vertex a has not yet saturated its ramp ($w_a(x) < 1$), so $\langle x, v_a\rangle < \tau_0 + \delta_w \ll \frac{\alpha}{2k+1}$, forcing $\langle G(x), x\rangle < \frac{3}{16}\alpha$ everywhere before the root r_k activates.
   - Second, on the positive side ($\frac{1}{2}\mathbb{E}[\langle G(X_{\mathrm{act}}), f_{\mathcal{V}}(X'_{\mathrm{act}})\rangle + \langle G(X'_{\mathrm{act}}), f_{\mathcal{V}}(X_{\mathrm{act}})\rangle]$), the active sets $B(s')$ decompose each post-order prefix $\{u \leq_{\mathrm{post}} s'\} = \bigsqcup_{b \in B(s')} \mathrm{Subtree}(b)$ into disjoint subtrees, which implies the exact upper-triangular inner-product identity $|\mathrm{Path}(s, r_k) \cap B(s')| = \mathbb{I}[s \leq_{\mathrm{post}} s']$. Symmetrizing across any pair $(s, s') \in \mathcal{F}_k^2$ gives $\mathbb{I}[s \leq_{\mathrm{post}} s'] + \mathbb{I}[s' \leq_{\mathrm{post}} s] \geq 1$, guaranteeing a positive inner product $> \frac{31}{64}\alpha$ for *every* pair of pre-terminal states (x, x')—even if an equilibrium mixes over exponentially many steps of the trajectory!

2 Preliminaries

Notation 1 (Vectors, Projections, and Finite Distributions)**.** Throughout, let $\mathbb{N} := \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 := \{0, 1, 2, \dots\}$. For a finite set S, $|S|$ denotes its cardinality and $\sqcup$ denotes disjoint set union ($A \cap B = \emptyset$). Sequence concatenation is written $(a_1, \dots, a_p) \parallel (b_1, \dots, b_q) := (a_1, \dots, a_p, b_1, \dots, b_q)$, and $\mathbb{I}[P] \in \{0, 1\}$ denotes the indicator of predicate P. For $m \in \mathbb{N}_0$, $\mathbb{R}^m$ denotes m-dimensional Euclidean space (with $\mathbb{R}^0 := \{()\}$, $\mathbf{0}_0 := ()$, and $\|\mathbf{0}_0\|_2 := 0$), equipped with standard inner product $\langle x, y\rangle = \sum_{j=1}^m x_j y_j$, Euclidean norm $\|x\|_2 = \sqrt{\langle x, x\rangle}$, and ℓ_∞ norm $\|x\|_\infty = \max_{1 \leq j \leq m} |x_j|$. We write $B_m(z_0, r) := \{z \in \mathbb{R}^m : \|z - z_0\|_2 \leq r\}$ for the closed Euclidean ball of radius $r > 0$ centered at z_0, $S^{m-1} := \{z \in \mathbb{R}^m : \|z\|_2 = 1\}$ for the unit sphere, and

$$\Pi_{B_m(\mathbf{0},1)}(z) \;:=\; \frac{z}{\max\{1,\, \|z\|_2\}} \qquad \big(z \in \mathbb{R}^m\big)$$

for radial clamping onto the unit Euclidean ball. For scalars $t \in \mathbb{R}$ and $a < b$, let $(t)_+ := \max\{0, t\}$ and $\mathrm{clip}_{[a,b]}(t) := \min\{b, \max\{a, t\}\}$. When taking extrema over subsets $A \subseteq S$ of a family $w \in [0, 1]^S$, we adopt the standard empty-set conventions $\min_{b \in \emptyset} w_b := 1$ and $\max_{b \in \emptyset} w_b := -\infty$ (with $\max\{x, -\infty\} := x$ for all $x \in \mathbb{R}$), under which $\min_{b \in A \cup \{u\}} w_b = \min\big\{\min_{b \in A} w_b,\, w_u\big\}$ holds for all $A \subseteq S$ and $u \in S$.

For any nonempty set S, $\Delta_{\mathrm{fin}}(S)$ denotes the family of finitely supported probability distributions $\mu = \sum_{r=1}^M p_r \delta_{z^{(r)}}$ ($M \in \mathbb{N}$, $p_r > 0$, $\sum_{r=1}^M p_r = 1$, $z^{(r)} \in S$), with finite support $\mathrm{supp}(\mu) := \{z \in S : \mu(\{z\}) > 0\}$ and uniform distribution $\mathrm{Unif}(S) := \sum_{z \in S} \frac{1}{|S|}\delta_z$ when S is finite. For any $\nu \in \Delta_{\mathrm{fin}}(S)$ and functions $F, H : S \to \mathbb{R}^m$, the Hilbert space $L^2(\nu; \mathbb{R}^m)$ has inner product $\langle F, H\rangle_{L^2(\nu;\mathbb{R}^m)} := \mathbb{E}_{Z \sim \nu}[\langle F(Z), H(Z)\rangle]$ and norm $\|F\|_{L^2(\nu;\mathbb{R}^m)} := \sqrt{\langle F, F\rangle_{L^2(\nu;\mathbb{R}^m)}}$.

Definition 1 (Two-Player Convex Games and ϵ-Approximate Correlated Equilibria)**.** A *two-player convex game* $G = (\mathcal{X}_1, \mathcal{X}_2, c_1, c_2)$ specifies (writing $-i := 3 - i$ and $(x, y) := (x_1, x_2) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$):

1. Compact convex strategy spaces $\mathcal{X}_i \subset \mathbb{R}^{d_i}$ ($i \in \{1, 2\}$, $d := d_1 + d_2$) satisfying $B_{d_i}(\mathbf{0}, 1) \subseteq \mathcal{X}_i \subseteq B_{d_i}(\mathbf{0}, R)$ for some radius $R \geq 1$;

2. Cost functions $c_i : \mathcal{X}_1 \times \mathcal{X}_2 \to [0, 1]$ ($i \in \{1, 2\}$) such that:

   - **(Own convexity)** for every fixed $x_{-i} \in \mathcal{X}_{-i}$, $x_i \mapsto c_i(x_i, x_{-i})$ is convex on $\mathcal{X}_i$; when differentiable in x_i, we write $\nabla_x c_1(x, y) := \nabla_{x_1} c_1(x, y) \in \mathbb{R}^{d_1}$ and $\nabla_y c_2(x, y) := \nabla_{x_2} c_2(x, y) \in \mathbb{R}^{d_2}$ for player i's own partial gradient and $\nabla_x^2 c_1(x, y), \nabla_y^2 c_2(x, y)$ for the own partial Hessians;
   - **(Euclidean L-Lipschitzness)** for all $x_1, x_1' \in \mathcal{X}_1$ and $x_2, x_2' \in \mathcal{X}_2$,

$$\big|c_i(x_1, x_2) - c_i(x_1', x_2')\big| \;\leq\; L\Big(\|x_1 - x_1'\|_2 + \|x_2 - x_2'\|_2\Big).$$

For any correlated distribution $\mu \in \Delta_{\text{fin}}(\mathcal{X}_1 \times \mathcal{X}_2)$, the *swap gap* (maximum expected swap regret) of player $i \in \{1, 2\}$ is

$$\text{SwapGap}_i(\mu) \;:=\; \sup_{\phi_i : \mathcal{X}_i \to \mathcal{X}_i} \mathbb{E}_{(X_1, X_2) \sim \mu}\Big[c_i(X_i, X_{-i}) - c_i\big(\phi_i(X_i), X_{-i}\big)\Big],$$

where ϕ_i ranges over all functions from $\mathcal{X}_i$ to $\mathcal{X}_i$. A distribution $\mu \in \Delta_{\text{fin}}(\mathcal{X}_1 \times \mathcal{X}_2)$ is an *ϵ-approximate normal-form correlated equilibrium (ϵ-CE)* if $\max\{\text{SwapGap}_1(\mu), \text{SwapGap}_2(\mu)\} \leq \epsilon$.

Definition 2 (Black-Box Oracle Model (M1))**.** In the black-box model **(M1)** [1] (Section 2.1), an algorithm is given (d_1, d_2, R, L) and accesses:

1. A *strong separation oracle* $\text{SEP}_{\mathcal{X}_i}$ for each $\mathcal{X}_i$ ($i \in \{1, 2\}$) which, on input $z \in \mathbb{R}^{d_i}$, returns `"FEASIBLE"` if $z \in \mathcal{X}_i$, or a vector $h \in \mathbb{R}^{d_i}$ with $\|h\|_\infty = 1$ and $\max_{y \in \mathcal{X}_i} \langle h, y \rangle < \langle h, z \rangle$ if $z \notin \mathcal{X}_i$;

2. A *first-order game oracle* $\mathcal{O}_G$ which, on any query $(x_1, x_2) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$, returns

$$\mathcal{O}_G(x_1, x_2) \;:=\; \big(c_1(x_1, x_2),\, c_2(x_1, x_2),\, \nabla_{x_1} c_1(x_1, x_2),\, \nabla_{x_2} c_2(x_1, x_2)\big).$$

After at most Q_0 oracle queries and unrestricted internal computation, the algorithm outputs a finitely supported distribution $\mu = \sum_{j=1}^{M} p_j \delta_{(x_1^{(j)}, x_2^{(j)})} \in \Delta_{\text{fin}}(\mathcal{X}_1 \times \mathcal{X}_2)$ of arbitrary support size $M \in \mathbb{N}$.

External Result 1 (Yao's Minimax Principle [8], Theorem 3)**.** *For any finite nonempty instance family $\mathcal{I}$, distribution $\pi_0 \in \Delta_{\text{fin}}(\mathcal{I})$, query budget $Q_0 \in \mathbb{N}_0$, randomized Q_0-query algorithm $\mathcal{R}$ (with internal randomness $\xi \sim \nu$ selecting deterministic Q_0-query algorithms $\mathcal{A}_\xi$), and binary predicate $\mathcal{S}(I, \mu_{\text{out}}) \in \{0, 1\}$,*

$$\inf_{I \in \mathcal{I}} \Pr_{\xi \sim \nu}\Big[\mathcal{S}\big(I, \mathcal{A}_\xi(I)\big) = 1\Big] \;\leq\; \sup_{\mathcal{A}\ \text{det.},\, \leq Q_0} \Pr_{I \sim \pi_0}\Big[\mathcal{S}\big(I, \mathcal{A}(I)\big) = 1\Big].$$

3 Combinatorial Gadget: The Post-Order Tree $\mathcal{F}_k$

To trace an exponential-length trajectory $s_1, \dots, s_N$ while storing at most $k+1$ active coordinates at any moment *and* maintaining an exact partition of unity across all continuous intermediate states $w \in [0,1]^{\mathcal{F}_k}$, we build a rooted tree $\mathcal{F}_k$ of depth parameter k. Each recursive step joins a left copy $\mathcal{F}_{\ell-1}^{(L)}$ and a right copy $\mathcal{F}_{\ell-1}^{(R)}$ using a unary *bridge vertex* b_ℓ (parent of $r_{\ell-1}^{(L)}$) followed by the binary root r_ℓ (parent of b_ℓ and $r_{\ell-1}^{(R)}$). Inserting b_ℓ ensures that every vertex transition introduces or retires at most one subtree ceiling at a time so that the activation weights telescope identically.

Definition 3 (Post-Order Binary-Unary Tree $\mathcal{F}_k$, Active Sets, and Partition Functions)**.** For any integer $\ell \in \mathbb{N}_0$ and disjoint ambient vertex sets $\mathrm{AmbReq}, \mathrm{AmbBlock}$, define recursively a rooted tree $\mathcal{F}_\ell = (V(\mathcal{F}_\ell), \mathrm{children})$ with root r_ℓ, ordered post-order list $\mathrm{PostOrder}(\mathcal{F}_\ell)$, ancestor paths $\mathrm{Path}_{\mathcal{F}_\ell}(u, r_\ell)$, and vertex subsets $\mathrm{Req}(u), \mathrm{Block}(u)$ for each $u \in V(\mathcal{F}_\ell)$ on fresh pairwise disjoint vertices:

- **Base case ($\ell = 0$):** Create a single leaf r_0 disjoint from $\mathrm{AmbReq} \sqcup \mathrm{AmbBlock}$ with $V(\mathcal{F}_0) := \{r_0\}$, $\mathrm{children}(r_0) := \emptyset$, $\mathrm{PostOrder}(\mathcal{F}_0) := (r_0)$, $\mathrm{Path}_{\mathcal{F}_0}(r_0, r_0) := \{r_0\}$, $\mathrm{Req}(r_0) := \mathrm{AmbReq}$, and $\mathrm{Block}(r_0) := \{r_0\} \sqcup \mathrm{AmbBlock}$.
- **Inductive step ($\ell \geq 1$):** Create two distinct fresh vertices b_ℓ (*bridge*) and r_ℓ (*root*) disjoint from $\mathrm{AmbReq} \sqcup \mathrm{AmbBlock}$, and:
  1. Build left copy $\mathcal{F}_{\ell-1}^{(L)}$ (root $r_{\ell-1}^{(L)}$) with $\mathrm{AmbReq}_L := \mathrm{AmbReq}$ and $\mathrm{AmbBlock}_L := \{b_\ell, r_\ell\} \sqcup \mathrm{AmbBlock}$;
  2. Set $\mathrm{children}(b_\ell) := \{r_{\ell-1}^{(L)}\}$, $\mathrm{Req}(b_\ell) := \mathrm{AmbReq} \sqcup \{r_{\ell-1}^{(L)}\}$, and $\mathrm{Block}(b_\ell) := \{b_\ell, r_\ell\} \sqcup \mathrm{AmbBlock}$;
  3. Recursively build right subtree $\mathcal{F}_{\ell-1}^{(R)}$ with root $r_{\ell-1}^{(R)}$, $\mathrm{AmbReq}_R := \mathrm{AmbReq} \sqcup \{b_\ell\}$, and $\mathrm{AmbBlock}_R := \{r_\ell\} \sqcup \mathrm{AmbBlock}$;
  4. Set $\mathrm{children}(r_\ell) := \{b_\ell, r_{\ell-1}^{(R)}\}$, $\mathrm{Req}(r_\ell) := \mathrm{AmbReq} \sqcup \{b_\ell, r_{\ell-1}^{(R)}\}$, and $\mathrm{Block}(r_\ell) := \{r_\ell\} \sqcup \mathrm{AmbBlock}$, keeping $(\mathrm{children}(u), \mathrm{Req}(u), \mathrm{Block}(u))$ unchanged for all $u \in V(\mathcal{F}_{\ell-1}^{(L)}) \sqcup V(\mathcal{F}_{\ell-1}^{(R)})$;
  5. Set $V(\mathcal{F}_\ell) := V(\mathcal{F}_{\ell-1}^{(L)}) \sqcup \{b_\ell\} \sqcup V(\mathcal{F}_{\ell-1}^{(R)}) \sqcup \{r_\ell\}$ and

$$\mathrm{PostOrder}(\mathcal{F}_\ell) \;:=\; \mathrm{PostOrder}(\mathcal{F}_{\ell-1}^{(L)}) \parallel (b_\ell) \parallel \mathrm{PostOrder}(\mathcal{F}_{\ell-1}^{(R)}) \parallel (r_\ell).$$

Along $\mathrm{PostOrder}(\mathcal{F}_\ell) = (u_1, \dots, u_{N_\ell})$ ($N_\ell := |V(\mathcal{F}_\ell)|$), let $\mathrm{first}(\mathcal{F}_\ell) := u_1$, write $u_i <_{\mathrm{post}} u_j \iff i < j$ and $u_i \leq_{\mathrm{post}} u_j \iff i \leq j$, define $\mathrm{Path}_{\mathcal{F}_\ell}(r_\ell, r_\ell) := \{r_\ell\}$ and $\mathrm{Path}_{\mathcal{F}_\ell}(c, r_\ell) := \{c\} \sqcup \mathrm{Path}_{\mathcal{F}_\ell}(u, r_\ell)$ for each $c \in \mathrm{children}(u)$, and define for each $u \in V(\mathcal{F}_\ell)$ its *next-step active set*

$$B(u) \;:=\; \Big(\mathrm{Req}(u) \setminus \mathrm{children}(u)\Big) \;\cup\; \{u\}.$$

Instantiating $\ell = k \geq 2$ with $\mathrm{AmbReq} = \mathrm{AmbBlock} = \emptyset$ defines the top-level tree $\mathcal{F}_k$ of size $N := |\mathcal{F}_k|$ with post-order sequence $(s_1, \dots, s_N)$, root $r_k = s_N$, root paths $\mathrm{Path}(s, r_k) := \mathrm{Path}_{\mathcal{F}_k}(s, r_k)$, and subtrees $\mathrm{Subtree}(a) := \{s \in \mathcal{F}_k : a \in \mathrm{Path}(s, r_k)\}$. For any activation vector $w = (w_a)_{a \in \mathcal{F}_k} \in [0,1]^{\mathcal{F}_k}$ and any $s, a \in \mathcal{F}_k$, define

$$L_s(w) \;:=\; \left(\min_{b \in \mathrm{Req}(s)} w_b \;-\; \max_{b \in \mathrm{Block}(s)} w_b\right)_+, \qquad S_a(w) \;:=\; \sum_{s \in \mathrm{Subtree}(a)} L_s(w).$$

Our first lemma establishes the basic recursive properties of $\mathcal{F}_k$: each vertex s_t's next-step active set $B(s_t)$ becomes the prerequisite set $\mathrm{Req}(s_{t+1})$ of the very next post-order vertex, has size at most $k+1$, and yields an exact partition of unity $\sum_{s\in\mathcal{F}_k} L_s(w) = 1 - w_{r_k}$.

Lemma 1 (Tree Invariants, Telescoping Partition of Unity, and Subtree Complementarity)**.** *For every integer $\ell \in \{0,1,\dots,k\}$:*

1. ***(Sizes and post-order transitions):*** *Whenever $\mathcal{F}_\ell$ is instantiated with disjoint sets* $\mathrm{AmbReq}, \mathrm{AmbBlock}$ *disjoint from $V(\mathcal{F}_\ell)$, we have $|V(\mathcal{F}_\ell)| = 3\cdot 2^\ell - 2$, and every $s \in V(\mathcal{F}_\ell)$ satisfies:*

$$\begin{aligned} \mathrm{Block}(s) &= \mathrm{Path}_{\mathcal{F}_\ell}(s, r_\ell) \sqcup \mathrm{AmbBlock}, \quad 1 \le |\mathrm{Path}_{\mathcal{F}_\ell}(s, r_\ell)| \le 2\ell + 1, \\ \mathrm{Req}(s) \cap \mathrm{Block}(s) &= \emptyset, \qquad\qquad |\mathrm{Req}(s)|, |B(s)| \le |\mathrm{AmbReq}| + \ell + 1. \end{aligned}$$

   Moreover, along $\mathrm{PostOrder}(\mathcal{F}_\ell) = (u_1, \dots, u_{N_\ell})$*, we have* $\mathrm{Req}(\mathrm{first}(\mathcal{F}_\ell)) = \mathrm{AmbReq}$*, $B(u_{N_\ell}) = B(r_\ell) = \mathrm{AmbReq} \sqcup \{r_\ell\}$, and $B(u_t) = \mathrm{Req}(u_{t+1})$ for all $1 \le t \le N_\ell - 1$. In particular, in $\mathcal{F}_k$ (*$\mathrm{AmbReq} = \mathrm{AmbBlock} = \emptyset$*), every $t \in \{1,\dots,N\}$ satisfies $|B(s_t)| \le k+1$,* $\mathrm{Req}(s_t) \subseteq \{s_1, \dots, s_{t-1}\}$*, and $B(s_t) \subseteq \{s_1, \dots, s_t\}$.*

2. ***(Telescoping partition of unity on*** $[0,1]^{\mathcal{F}_k}$***):*** *For every $w \in [0,1]^{\mathcal{F}_k}$, writing $q := \min_{a\in\mathrm{AmbReq}} w_a \in [0,1]$ and $f := \max_{b\in\mathrm{AmbBlock}} w_b \in [0,1] \cup \{-\infty\}$,*

$$\sum_{s\in V(\mathcal{F}_\ell)} L_s(w) = \big(q - \max\{w_{r_\ell}, f\}\big)_+.$$

   At $\ell = k$ ($\mathrm{AmbReq} = \mathrm{AmbBlock} = \emptyset$*, so $q = 1$ and $f = -\infty$), $\sum_{s\in\mathcal{F}_k} L_s(w) = 1 - w_{r_k}$ for all $w \in [0,1]^{\mathcal{F}_k}$.*

3. ***(Ancestor complementarity):*** *For every vertex $a \in \mathcal{F}_k$ and every $w \in [0,1]^{\mathcal{F}_k}$,*

$$0 \le S_a(w) \le 1 - w_a \qquad \text{and} \qquad \sum_{b\in\mathcal{F}_k} S_b(w) \le (2k+1) \sum_{s\in\mathcal{F}_k} L_s(w).$$

Proof. Since Definition 3 preserves $(\mathrm{children}(u), \mathrm{Req}(u), \mathrm{Block}(u))$ across recursive calls, all vertex attributes agree with those in $\mathcal{F}_k$.

(1) We induct on $\ell \in \{0,\dots,k\}$. For $\ell = 0$, $V(\mathcal{F}_0) = \{r_0\}$ has size $1 = 3\cdot 2^0 - 2$, $\mathrm{Path}_{\mathcal{F}_0}(r_0, r_0) = \{r_0\}$ of size 1, $\mathrm{Block}(r_0) = \{r_0\} \sqcup \mathrm{AmbBlock}$, $\mathrm{Req}(r_0) = \mathrm{AmbReq}$ disjoint from $\mathrm{Block}(r_0)$, and $B(r_0) = \mathrm{AmbReq} \sqcup \{r_0\}$ of size $|\mathrm{AmbReq}| + 1$.

For $\ell \ge 1$, $|V(\mathcal{F}_\ell)| = 2(3\cdot 2^{\ell-1} - 2) + 2 = 3\cdot 2^\ell - 2$. Checking each block of $V(\mathcal{F}_\ell) = V(\mathcal{F}_{\ell-1}^{(L)}) \sqcup \{b_\ell\} \sqcup V(\mathcal{F}_{\ell-1}^{(R)}) \sqcup \{r_\ell\}$:

- On $V(\mathcal{F}_{\ell-1}^{(L)})$ ($\mathrm{AmbReq}_L = \mathrm{AmbReq}$, $\mathrm{AmbBlock}_L = \{b_\ell, r_\ell\} \sqcup \mathrm{AmbBlock}$), the induction hypothesis gives $\mathrm{Path}_{\mathcal{F}_\ell}(s, r_\ell) = \mathrm{Path}_L(s, r_{\ell-1}^{(L)}) \sqcup \{b_\ell, r_\ell\}$ of size at most $(2\ell - 1) + 2 = 2\ell + 1$, $\mathrm{Block}(s) = \mathrm{Path}_{\mathcal{F}_\ell}(s, r_\ell) \sqcup \mathrm{AmbBlock}$ disjoint from $\mathrm{Req}(s)$, and $|\mathrm{Req}(s)|, |B(s)| \le |\mathrm{AmbReq}| + \ell$.

- At $s = b_\ell$, $\mathrm{Path}_{\mathcal{F}_\ell}(b_\ell, r_\ell) = \{b_\ell, r_\ell\}$ has size $2 \le 2\ell + 1$, $\mathrm{Req}(b_\ell) = \mathrm{AmbReq} \sqcup \{r_{\ell-1}^{(L)}\}$ is disjoint from $\mathrm{Block}(b_\ell) = \{b_\ell, r_\ell\} \sqcup \mathrm{AmbBlock}$, and $B(b_\ell) = \mathrm{AmbReq} \sqcup \{b_\ell\}$ has size $|\mathrm{AmbReq}| + 1$.

- On $V(\mathcal{F}_{\ell-1}^{(R)})$ ($\mathrm{AmbReq}_R = \mathrm{AmbReq} \sqcup \{b_\ell\}$, $\mathrm{AmbBlock}_R = \{r_\ell\} \sqcup \mathrm{AmbBlock}$), the induction hypothesis gives $\mathrm{Path}_{\mathcal{F}_\ell}(s, r_\ell) = \mathrm{Path}_R(s, r_{\ell-1}^{(R)}) \sqcup \{r_\ell\}$ of size at most 2ℓ, $\mathrm{Block}(s) = \mathrm{Path}_{\mathcal{F}_\ell}(s, r_\ell) \sqcup \mathrm{AmbBlock}$ disjoint from $\mathrm{Req}(s)$, and $|\mathrm{Req}(s)|, |B(s)| \le |\mathrm{AmbReq}_R| + \ell = |\mathrm{AmbReq}| + \ell + 1$.
- At $s = r_\ell$, $\mathrm{Path}_{\mathcal{F}_\ell}(r_\ell, r_\ell) = \{r_\ell\}$, $\mathrm{Req}(r_\ell) = \mathrm{AmbReq} \sqcup \{b_\ell, r_{\ell-1}^{(R)}\}$ has size $|\mathrm{AmbReq}| + 2 \le |\mathrm{AmbReq}| + \ell + 1$ and is disjoint from $\mathrm{Block}(r_\ell) = \{r_\ell\} \sqcup \mathrm{AmbBlock}$, and $B(r_\ell) = \mathrm{AmbReq} \sqcup \{r_\ell\}$ has size $|\mathrm{AmbReq}| + 1$.

Along $\mathrm{PostOrder}(\mathcal{F}_\ell)$, the induction hypothesis gives $\mathrm{Req}(\mathrm{first}(\mathcal{F}_\ell)) = \mathrm{AmbReq}_L = \mathrm{AmbReq}$ and handles transitions inside $\mathcal{F}_{\ell-1}^{(L)}$ and $\mathcal{F}_{\ell-1}^{(R)}$, while the three junction transitions match identically:

$$\begin{aligned} B(r_{\ell-1}^{(L)}) &= \mathrm{AmbReq} \sqcup \{r_{\ell-1}^{(L)}\} = \mathrm{Req}(b_\ell), \\ B(b_\ell) &= \mathrm{AmbReq} \sqcup \{b_\ell\} = \mathrm{Req}(\mathrm{first}(\mathcal{F}_{\ell-1}^{(R)})), \\ B(r_{\ell-1}^{(R)}) &= \mathrm{AmbReq} \sqcup \{b_\ell, r_{\ell-1}^{(R)}\} = \mathrm{Req}(r_\ell). \end{aligned}$$

Setting $\ell = k$ gives $\mathrm{Req}(s_1) = \emptyset$, and induction on $t = 1, \dots, N$ immediately gives $\mathrm{Req}(s_t) \subseteq \{s_1, \dots, s_{t-1}\}$ and $B(s_t) \subseteq \{s_1, \dots, s_t\}$.

(2) Notice first that for any real numbers $Q, A, F_0 \in \mathbb{R}$, exhaustively checking $A \le F_0$, $F_0 < A < Q$, and $A \ge Q$ gives the elementary telescoping identity

$$\big(Q - \max\{A, F_0\}\big)_+ + \big(\min\{Q, A\} - F_0\big)_+ = \big(Q - F_0\big)_+.$$

We induct on $\ell \in \{0, \dots, k\}$. For $\ell = 0$, $L_{r_0}(w) = (q - \max\{w_{r_0}, f\})_+$. For $\ell \ge 1$, let $f_R := \max\{w_{r_\ell}, f\} \in [0,1]$, $f_L := \max\{w_{b_\ell}, f_R\} \in [0,1]$, and $q_R := \min\{q, w_{b_\ell}\} \in [0,1]$. Applying the induction hypothesis and the telescoping identity above to $\mathcal{F}_{\ell-1}^{(L)} \sqcup \{b_\ell\}$ (eliminating $w_{r_{\ell-1}^{(L)}}$) and to $\mathcal{F}_{\ell-1}^{(R)} \sqcup \{r_\ell\}$ (eliminating $w_{r_{\ell-1}^{(R)}}$), and then summing the two results (eliminating w_{b_ℓ}) yields

$$\begin{aligned} \sum_{s \in V(\mathcal{F}_\ell)} L_s(w) &= \Big[\big(q - \max\{w_{r_{\ell-1}^{(L)}}, f_L\}\big)_+ + \big(\min\{q, w_{r_{\ell-1}^{(L)}}\} - f_L\big)_+\Big] \\ &\quad + \Big[\big(q_R - \max\{w_{r_{\ell-1}^{(R)}}, f_R\}\big)_+ + \big(\min\{q_R, w_{r_{\ell-1}^{(R)}}\} - f_R\big)_+\Big] \\ &= \big(q - \max\{w_{b_\ell}, f_R\}\big)_+ + \big(\min\{q, w_{b_\ell}\} - f_R\big)_+ = \big(q - f_R\big)_+. \end{aligned}$$

(3) Every vertex $a \in \mathcal{F}_k$ is introduced in Definition 3 either as a leaf root r_0 ($\ell = 0$, where $\mathrm{Subtree}(r_0) = \{r_0\}$ and $S_{r_0}(w) = L_{r_0}(w) \le 1 - w_{r_0}$), as a bridge b_ℓ ($\ell \ge 1$, where $\mathrm{Subtree}(b_\ell) = V(\mathcal{F}_{\ell-1}^{(L)}) \sqcup \{b_\ell\}$ has sum $(q - f_L)_+ \le 1 - w_{b_\ell}$), or as an internal root r_ℓ ($\ell \ge 1$, where $\mathrm{Subtree}(r_\ell) = V(\mathcal{F}_\ell)$ has sum $(q - f_R)_+ \le 1 - w_{r_\ell}$). Hence $0 \le S_a(w) \le 1 - w_a$ for every $a \in \mathcal{F}_k$. Finally, since $|\mathrm{Path}(s, r_k)| \le 2k + 1$ by (1), interchanging sums gives $\sum_{b \in \mathcal{F}_k} S_b(w) = \sum_{s \in \mathcal{F}_k} |\mathrm{Path}(s, r_k)| L_s(w) \le (2k+1) \sum_{s \in \mathcal{F}_k} L_s(w)$. □

Next, we formalize two structural consequences of the tree decomposition: at most one vertex per contiguous region between nonzero coordinates of w can have $L_s(w) > 0$, and no vertex after s_q can activate if $w_a = 0$ on $\{s_q, \dots, s_N\}$.

Lemma 2 (Lowest-Common-Ancestor Sparsity and Prefix Blocking)**.**

1. ***(Active node sparsity):*** *For every* $w \in [0,1]^{\mathcal{F}_k}$,

$$\big|\{s \in \mathcal{F}_k : L_s(w) > 0\}\big| \;\le\; 1 + \big|\{a \in \mathcal{F}_k \setminus \{r_k\} : w_a > 0\}\big|.$$

2. ***(Prefix blocking):*** *For every* $q \in \{1,\dots,N-1\}$ *and every vertex* $b \in \{s_{q+1},\dots,s_N\}$,

$$\mathrm{Req}(b) \;\cap\; \{s_q,\dots,s_N\} \;\neq\; \emptyset.$$

Proof. For any pair $u <_{\mathrm{post}} v$ in $\mathcal{F}_k$, let $\mathcal{F}_\ell$ ($\ell \ge 1$) be the smallest recursive tree containing both u and v. Across $V(\mathcal{F}_\ell) = V(\mathcal{F}_{\ell-1}^{(L)}) \,\|\, \{b_\ell\} \,\|\, V(\mathcal{F}_{\ell-1}^{(R)}) \,\|\, \{r_\ell\}$, minimality excludes both $(u,v) \in V(\mathcal{F}_{\ell-1}^{(L)})^2$ and $(u,v) \in V(\mathcal{F}_{\ell-1}^{(R)})^2$, leaving three exhaustive cases for a separator vertex $u^\star(u,v) \in \mathcal{F}_k \setminus \{r_k\}$:

- if $u \in V(\mathcal{F}_{\ell-1}^{(L)})$ and $v = b_\ell$, set $u^\star(u,v) := r_{\ell-1}^{(L)}$ ($\mathrm{Subtree}(u^\star) = V(\mathcal{F}_{\ell-1}^{(L)})$, $u^\star \in \mathrm{Req}(b_\ell)$);
- if $u \in V(\mathcal{F}_{\ell-1}^{(L)}) \sqcup \{b_\ell\}$ and $v \in V(\mathcal{F}_{\ell-1}^{(R)}) \sqcup \{r_\ell\}$, set $u^\star(u,v) := b_\ell$ ($\mathrm{Subtree}(u^\star) = V(\mathcal{F}_{\ell-1}^{(L)}) \sqcup \{b_\ell\}$, $u^\star \in \mathrm{AmbReq}_R \subseteq \mathrm{Req}(v)$);
- if $u \in V(\mathcal{F}_{\ell-1}^{(R)})$ and $v = r_\ell$, set $u^\star(u,v) := r_{\ell-1}^{(R)}$ ($\mathrm{Subtree}(u^\star) = V(\mathcal{F}_{\ell-1}^{(R)})$, $u^\star \in \mathrm{Req}(r_\ell)$).

In all three cases, $\mathrm{Subtree}(u^\star(u,v))$ is a contiguous post-order interval containing u and ending at $u^\star(u,v)$, and $u^\star(u,v) \in \mathrm{Req}(v)$ satisfies $u \le_{\mathrm{post}} u^\star(u,v) <_{\mathrm{post}} v$.

For (1), list $\{s \in \mathcal{F}_k : L_s(w) > 0\}$ in post-order as $s_{i_1} <_{\mathrm{post}} \cdots <_{\mathrm{post}} s_{i_p}$ ($p \ge 2$). Each $a_j := u^\star(s_{i_{j-1}}, s_{i_j}) \in \mathcal{F}_k \setminus \{r_k\}$ ($2 \le j \le p$) belongs to $\mathrm{Req}(s_{i_j})$, so $w_{a_j} \ge L_{s_{i_j}}(w) > 0$. Moreover, for $2 \le j < j' \le p$, $\mathrm{Subtree}(a_j)$ ends at $a_j <_{\mathrm{post}} s_{i_j} \le_{\mathrm{post}} s_{i_{j'-1}}$, so $s_{i_{j'-1}} \notin \mathrm{Subtree}(a_j)$ whereas $s_{i_{j'-1}} \in \mathrm{Subtree}(a_{j'})$. Hence $a_2,\dots,a_p$ are $p-1$ distinct nonzero non-root coordinates of w.

For (2), since $s_q <_{\mathrm{post}} b$ for every $b \in \{s_{q+1},\dots,s_N\}$, setting $a_b := u^\star(s_q, b)$ gives $a_b \in \mathrm{Req}(b)$ and $s_q \le_{\mathrm{post}} a_b$, so $a_b \in \mathrm{Req}(b) \cap \{s_q,\dots,s_N\}$. □

Finally, we prove the key combinatorial property of the active stack $B(s')$: for every vertex $s' \in \mathcal{F}_k$, the subtrees rooted at $B(s')$ partition the post-order prefix $\{u \in \mathcal{F}_k : u \le_{\mathrm{post}} s'\}$, which immediately implies the upper-triangular path intersection matrix $|\mathrm{Path}(s, r_k) \cap B(s')| = \mathbb{I}[s \le_{\mathrm{post}} s']$.

Lemma 3 (Disjoint Prefix Partition and Upper-Triangular Path Intersection)**.** *For every vertex* $s' \in \mathcal{F}_k$*, the subtrees* $\{\mathrm{Subtree}(b) : b \in B(s')\}$ *are pairwise disjoint and satisfy*

$$\{u \in \mathcal{F}_k : u \le_{\mathrm{post}} s'\} \;=\; \bigsqcup_{b \in B(s')} \mathrm{Subtree}(b).$$

Consequently, for every ordered pair of vertices $(s, s') \in \mathcal{F}_k \times \mathcal{F}_k$,

$$\Big|\mathrm{Path}(s,\, r_k) \;\cap\; B(s')\Big| \;=\; \mathbb{I}\big[\, s \;\le_{\mathrm{post}}\; s' \,\big].$$

Proof. Write $B_0 := \emptyset$ and $B_t := B(s_t)$ for $t = 1,\dots,N$. By Lemma 1(1), $\mathrm{Req}(s_1) = B_0$ and $\mathrm{Req}(s_t) = B_{t-1}$ for all $2 \le t \le N$. Notice that every node $s_t \in \mathcal{F}_k$ is placed immediately after all subtrees of its children in $\mathrm{PostOrder}(\mathcal{F}_k)$, so $\mathrm{Subtree}(s_t) = \big(\bigsqcup_{c \in \mathrm{children}(s_t)} \mathrm{Subtree}(c)\big) \sqcup \{s_t\}$. We verify by a short induction on $t = 1,\dots,N$ that $\{s_1,\dots,s_t\} = \bigsqcup_{b \in B_t} \mathrm{Subtree}(b)$:

- For $t = 1$, s_1 is a leaf ($\mathrm{children}(s_1) = \emptyset$), $B_1 = \{s_1\}$, and $\mathrm{Subtree}(s_1) = \{s_1\}$.

- Given $\{s_1, \dots, s_{t-1}\} = \bigsqcup_{b \in B_{t-1}} \mathrm{Subtree}(b)$ with $B_{t-1} = \mathrm{Req}(s_t) \supseteq \mathrm{children}(s_t)$, removing $\mathrm{children}(s_t)$ from B_{t-1} and adding s_t replaces $\bigsqcup_{c \in \mathrm{children}(s_t)} \mathrm{Subtree}(c)$ together with $\{s_t\} \not\subseteq \{s_1, \dots, s_{t-1}\}$ by the single subtree $\mathrm{Subtree}(s_t)$, yielding the disjoint union $\{s_1, \dots, s_t\} = \bigsqcup_{b \in B_t} \mathrm{Subtree}(b)$.

Since $b \in \mathrm{Path}(s, r_k) \iff s \in \mathrm{Subtree}(b)$, the sets $\{\mathrm{Subtree}(b) : b \in B(s')\}$ being pairwise disjoint with union $\{u \in \mathcal{F}_k : u \leq_{\mathrm{post}} s'\}$ immediately gives $|\mathrm{Path}(s, r_k) \cap B(s')| = |\{b \in B(s') : s \in \mathrm{Subtree}(b)\}| = \mathbb{I}[s \leq_{\mathrm{post}} s']$. □

4 Hard Game Construction and Analytical Properties

Definition 4 (Hard Two-Player Game Instances $G_{\mathcal{V}}$ and $G_{\mathcal{V},d}$)**.** For any integer $k \geq 2$, define the fixed deterministic parameters:

$$N := 3 \cdot 2^k - 2, \qquad \alpha := \frac{1}{16(k+1)}, \qquad \delta_w := \frac{\alpha}{8(2k+1)},$$
$$\tau_0 := \frac{\delta_w}{16} = \frac{\alpha}{128(2k+1)}, \qquad \mu_{\mathrm{coh}} := \frac{\tau_0^2}{16}, \qquad d_0 := 10^{24}\, k^{16}.$$

Given any tuple of unit vectors $\mathcal{V} = (v_a)_{a \in \mathcal{F}_k} \in \left(S^{d_0-1}\right)^{\mathcal{F}_k}$ satisfying $|\langle v_a, v_b \rangle| \leq \mu_{\mathrm{coh}}$ for all distinct $a \neq b \in \mathcal{F}_k$:

1. **Radially clamped ramp activations:** For every $z \in \mathbb{R}^{d_0}$, let $\hat{z} := \Pi_{B_{d_0}(\mathbf{0},1)}(z) = \frac{z}{\max\{1, \|z\|_2\}} \in B_{d_0}(\mathbf{0}, 1)$, define inner products $t_a(\hat{z}) := \langle \hat{z}, v_a \rangle$ $(a \in \mathcal{F}_k)$, and define $w(\hat{z}) = (w_a(\hat{z}))_{a \in \mathcal{F}_k} \in [0,1]^{\mathcal{F}_k}$ by
$$w_a(\hat{z}) := \mathrm{clip}_{[0,1]}\left(\frac{\langle \hat{z},\, v_a \rangle - \tau_0}{\delta_w}\right).$$

2. **Post-order target self-map $f_{\mathcal{V}} : \mathbb{R}^{d_0} \to \mathbb{R}^{d_0}$:** For every $z \in \mathbb{R}^{d_0}$, define
$$f_{\mathcal{V}}(z) := \alpha\, w_{r_k}(\hat{z})\, v_{r_k} + \alpha \sum_{s \in \mathcal{F}_k} L_s\big(w(\hat{z})\big) \sum_{b \in B(s)} v_b.$$

3. **Active-space game $G_{\mathcal{V}}$ and unit-ball padded game $G_{\mathcal{V},d}$ $(R := 1)$:** Define active costs $c_1^{\mathrm{act}}, c_2^{\mathrm{act}} : \mathbb{R}^{d_0} \times \mathbb{R}^{d_0} \to [0, +\infty)$ by
$$c_1^{\mathrm{act}}(u, v) := \frac{1}{4}\|u - v\|_2^2, \qquad c_2^{\mathrm{act}}(u, v) := \frac{1}{4}\big\|v - f_{\mathcal{V}}(u)\big\|_2^2,$$
forming $G_{\mathcal{V}} := \big(B_{d_0}(\mathbf{0}, 1), B_{d_0}(\mathbf{0}, 1), c_1^{\mathrm{act}}, c_2^{\mathrm{act}}\big)$. For any ambient dimension $d \geq d_{\min} := 2 \cdot 10^{24} \cdot 2^{16}$, set $d_1 := \lfloor d/2 \rfloor$, $d_2 := \lceil d/2 \rceil$, $k(d) := \left\lfloor \left(\frac{d_1}{10^{24}}\right)^{1/16} \right\rfloor \geq 2$, and $d_0 := 10^{24} k(d)^{16} \leq d_1 \leq d_2$. On $\mathcal{X}_1 := B_{d_1}(\mathbf{0}, 1)$ and $\mathcal{X}_2 := B_{d_2}(\mathbf{0}, 1)$, decomposing $x = (x_{\mathrm{act}}, x_{\mathrm{pad}}) \in \mathbb{R}^{d_0} \times \mathbb{R}^{d_1 - d_0}$ and $y = (y_{\mathrm{act}}, y_{\mathrm{pad}}) \in \mathbb{R}^{d_0} \times \mathbb{R}^{d_2 - d_0}$, define
$$c_1(x, y) := c_1^{\mathrm{act}}(x_{\mathrm{act}}, y_{\mathrm{act}}) + \frac{1}{4}\|x_{\mathrm{pad}}\|_2^2, \qquad c_2(x, y) := c_2^{\mathrm{act}}(x_{\mathrm{act}}, y_{\mathrm{act}}) + \frac{1}{4}\|y_{\mathrm{pad}}\|_2^2,$$
defining the padded two-player convex game $G_{\mathcal{V},d} := (\mathcal{X}_1, \mathcal{X}_2, c_1, c_2)$.

We next establish standard Rademacher tail bounds, pairwise incoherence, and an elementary Cauchy–Schwarz active-sparsity bound showing that at most $O(\tau_0^{-2}) = O(k^4)$ directions can exceed threshold τ_0 at any point of $B_{d_0}(\mathbf{0}, 1)$.

Lemma 4 (Rademacher Concentration, Pairwise Incoherence, and Active Coordinate Sparsity)**.** *Let $k \geq 2$ be any integer.*

1. ***(Unit-ball sub-Gaussian tail):*** *For every deterministic vector $u \in B_{d_0}(\mathbf{0}, 1)$, uniform Rademacher unit vector $v \sim \mathrm{Unif}\big(\{\pm d_0^{-1/2}\}^{d_0}\big)$, and threshold $\tau > 0$,*

$$\Pr_v\Big[\big|\langle u,\, v\rangle\big| > \tau\Big] \;\leq\; 2\exp\!\left(-\frac{d_0\,\tau^2}{2}\right), \qquad \textit{and in particular} \qquad \Pr_v\Big[\big|\langle u,\, v\rangle\big| > \frac{\tau_0}{2}\Big] \;<\; 2^{-30k}.$$

2. ***(Pairwise incoherence):*** *For independent uniform Rademacher unit vectors $\mathcal{V} = (v_a)_{a\in\mathcal{F}_k} \sim \mathrm{Unif}\big(\{\pm d_0^{-1/2}\}^{d_0\times\mathcal{F}_k}\big)$, with probability at least $1 - 2^{-8k}$, $|\langle v_a, v_b\rangle| \leq \mu_{\mathrm{coh}} = \frac{\tau_0^2}{16}$ holds simultaneously for all distinct $a \neq b \in \mathcal{F}_k$.*

3. ***(Active coordinate sparsity):*** *For any μ_{coh}-incoherent unit vector tuple $\mathcal{V} = (v_a)_{a\in\mathcal{F}_k}$ and every $\hat{z} \in B_{d_0}(\mathbf{0}, 1)$,*

$$\big|A(\hat{z})\big| \;:=\; \big|\big\{a \in \mathcal{F}_k : |t_a(\hat{z})| > \tau_0\big\}\big| \;<\; \frac{16}{15\,\tau_0^2}, \qquad \textit{so} \qquad 1 + \big|A(\hat{z})\big| \;<\; \frac{2}{\tau_0^2} \;<\; 10^{10}\,k^6.$$

Proof. **(1)** Write $v = d_0^{-1/2}(\sigma_1, \ldots, \sigma_{d_0})$ with i.i.d. uniform signs $\sigma_j \sim \mathrm{Unif}\{-1, +1\}$. Since $\frac{e^{\eta}+e^{-\eta}}{2} = \sum_{m=0}^{\infty} \frac{\eta^{2m}}{(2m)!} \leq \exp(\eta^2/2)$ for every $\eta \in \mathbb{R}$, setting $\theta := d_0\tau > 0$ and using $\|u\|_2 \leq 1$ yields

$$\mathbb{E}_v\big[\exp(\theta\langle u, v\rangle)\big] \;=\; \prod_{j=1}^{d_0} \mathbb{E}_{\sigma_j}\left[\exp\!\left(\frac{\theta u_j}{\sqrt{d_0}}\,\sigma_j\right)\right] \;\leq\; \exp\!\left(\frac{\theta^2\|u\|_2^2}{2d_0}\right) \;\leq\; \exp\!\left(\frac{d_0\tau^2}{2}\right).$$

Applying Markov's inequality to $\pm u$ and union-bounding gives $\Pr_v[|\langle u, v\rangle| > \tau] \leq 2\exp(-d_0\tau^2/2)$. Since $k \geq 2$ implies $\frac{1}{\tau_0} = 2048(k+1)(2k+1) \leq 7680k^2$, setting $\tau = \frac{\tau_0}{2}$ yields $\frac{d_0(\tau_0/2)^2}{2} \geq \frac{10^{24}k^{16}}{8(7680k^2)^2} > 2\cdot 10^{15}k^{12} > 30k + 1$, giving $2\exp(-d_0\tau_0^2/8) < 2^{-30k}$.

(2) For each unordered pair $\{a, b\} \subset \mathcal{F}_k$, conditioning on v_a and applying (1) at $\tau = \mu_{\mathrm{coh}} = \frac{\tau_0^2}{16}$ bounds $\Pr[|\langle v_a, v_b\rangle| > \mu_{\mathrm{coh}}] \leq 2\exp(-d_0\tau_0^4/512) < 2\exp(-100k)$ since $\frac{d_0\tau_0^4}{512} \geq \frac{10^{24}k^{16}}{512(7680k^2)^4} > 5\cdot 10^5 k^8 > 100k$. Union-bounding over all $\binom{N}{2} < 5\cdot 2^{2k}$ pairs yields failure probability $< 10\cdot 2^{2k}\cdot 2^{-100k} < 2^{-8k}$.

(3) Let $A := A(\hat{z})$ and $m := |A|$. If $m \geq 1$, write $\xi_a := \langle \hat{z}, v_a\rangle$ $(a \in A)$, so $\sum_{a\in A}\xi_a^2 > m\tau_0^2 > 0$. By Cauchy–Schwarz ($\|\hat{z}\|_2 \leq 1$) and $2|\xi_a\xi_b| \leq \xi_a^2 + \xi_b^2$,

$$\left(\sum_{a\in A}\xi_a^2\right)^2 \;=\; \left\langle \hat{z},\, \sum_{a\in A}\xi_a v_a\right\rangle^2 \;\leq\; \sum_{a\in A}\xi_a^2 + \mu_{\mathrm{coh}}\sum_{\substack{a,b\in A\\ a\neq b}}\frac{\xi_a^2+\xi_b^2}{2} \;\leq\; \big(1 + m\,\mu_{\mathrm{coh}}\big)\sum_{a\in A}\xi_a^2.$$

Dividing by $\sum_{a\in A}\xi_a^2 > 0$ gives $m\tau_0^2 < 1 + \frac{m\tau_0^2}{16}$, i.e., $m < \frac{16}{15\tau_0^2}$ and $1 + m < \frac{2}{\tau_0^2} \leq 2(7680k^2)^2 < 10^{10}k^6$. □

Combining active sparsity with piecewise-affine telescoping along segments shows that every game instance $G_{\mathcal{V},d}$ satisfies all convex game and oracle regularity requirements with Lipschitz modulus $L < \sqrt{d}$.

Lemma 5 (Regularity, Strong Convexity, Lipschitz Modulus, and Separation Oracle)**.** *For every $d \geq d_{\min}$ and every μ_{coh}-incoherent unit vector tuple $\mathcal{V} = (v_a)_{a\in\mathcal{F}_k}$:*

1. *Radial clamping $\hat{z} = \Pi_{B_{d_0}(\mathbf{0},1)}(z)$ is 1-Lipschitz on $\mathbb{R}^{d_0}$ ($\|\hat{z} - \hat{z}'\|_2 \leq \|z - z'\|_2$) with $\hat{z} = z$ on $B_{d_0}(\mathbf{0}, 1)$, and $\|f_{\mathcal{V}}(z)\|_2 \leq \frac{1}{16}$ for all $z \in \mathbb{R}^{d_0}$.*
2. *On $\mathcal{X}_1 = B_{d_1}(\mathbf{0}, 1)$ and $\mathcal{X}_2 = B_{d_2}(\mathbf{0}, 1)$, the costs $c_1, c_2 : \mathcal{X}_1 \times \mathcal{X}_2 \to [0, 1]$ satisfy $0 \leq c_i(x, y) \leq 1$, have own partial gradients $\nabla_x c_1(x, y) = \frac{1}{2}(x_{\mathrm{act}} - y_{\mathrm{act}}, x_{\mathrm{pad}})$ and $\nabla_y c_2(x, y) = \frac{1}{2}(y_{\mathrm{act}} - f_{\mathcal{V}}(x_{\mathrm{act}}), y_{\mathrm{pad}})$ with constant own Hessians $\nabla_x^2 c_1 = \frac{1}{2} I_{d_1} \succ \mathbf{0}$ and $\nabla_y^2 c_2 = \frac{1}{2} I_{d_2} \succ \mathbf{0}$, and are Euclidean L-Lipschitz with $L := 10^{12}k^8 = d_0^{1/2} < d^{1/2}$.*
3. *Each $\mathcal{X}_i = B_{d_i}(\mathbf{0}, 1)$ admits a deterministic strong separation oracle $\mathrm{SEP}_{\mathcal{X}_i}$ independent of $\mathcal{V}$ using at most $6d_i - 1$ scalar operations over $\{+, -, \times, /, <, \max\}$.*

Proof. **(1)** Since $\Pi_{B_{d_0}(\mathbf{0},1)}$ is the Euclidean metric projection onto the closed convex set $B_{d_0}(\mathbf{0}, 1)$, standard obtuse-angle variational inequalities give $\|\hat{z} - \hat{z}'\|_2 \leq \|z - z'\|_2$ for all $z, z' \in \mathbb{R}^{d_0}$. By Lemma 1(1)–(2), $|B(s)| \leq k + 1$ and $\sum_{s\in\mathcal{F}_k} L_s(w(\hat{z})) = 1 - w_{r_k}(\hat{z})$, so

$$\big\|f_{\mathcal{V}}(z)\big\|_2 \;\leq\; \alpha\, w_{r_k}(\hat{z}) + \alpha(k+1)\big(1 - w_{r_k}(\hat{z})\big) \;\leq\; \alpha(k+1) \;=\; \frac{1}{16}.$$

(2) For $(x, y) \in B_{d_1}(\mathbf{0}, 1) \times B_{d_2}(\mathbf{0}, 1)$, expanding Euclidean squares and using $\|f_{\mathcal{V}}(x_{\mathrm{act}})\|_2 \leq \frac{1}{16}$ gives $0 \leq c_1(x, y) \leq \frac{1}{4}(1 + 2 + 1) = 1$ and $0 \leq c_2(x, y) \leq \frac{1}{4}(1 + \frac{1}{8} + \frac{1}{256}) < 1$, with constant own Hessians $\frac{1}{2} I_{d_i}$.

To bound $\mathrm{Lip}(f_{\mathcal{V}})$, let $x, x' \in \mathbb{R}^{d_0}$ and let $\hat{z}(t) := (1 - t)\hat{x} + t\hat{x}' \in B_{d_0}(\mathbf{0}, 1)$ ($t \in [0, 1]$). Since each ramp $t \mapsto w_a(\hat{z}(t))$ is piecewise affine with slope magnitude at most $\frac{1}{\delta_w}\|\hat{x}' - \hat{x}\|_2$, there exists a finite partition $0 = t_0 < t_1 < \cdots < t_r = 1$ such that on every subinterval $[t_{j-1}, t_j]$, every function $g_s(t) := L_s(w(\hat{z}(t))) \geq 0$ ($s \in \mathcal{F}_k$) and $h_{r_k}(t) := w_{r_k}(\hat{z}(t))$ is affine. Let $t_j^\star := \frac{t_{j-1}+t_j}{2} \in (t_{j-1}, t_j)$ be the midpoint of each subinterval. Whenever $g_s(t_j^\star) = 0$, nonnegativity and affinity on $[t_{j-1}, t_j]$ imply $g_s \equiv 0$ on $[t_{j-1}, t_j]$, so $g_s'(t_j^\star) = 0$. Whenever $g_s(t_j^\star) > 0$, we have $|g_s'(t_j^\star)| \leq \frac{2}{\delta_w}\|\hat{x}' - \hat{x}\|_2$, and by Lemma 2(1) and Lemma 4(3), at most $1 + |A(\hat{z}(t_j^\star))| < 10^{10}k^6$ indices $s \in \mathcal{F}_k$ satisfy $g_s(t_j^\star) > 0$. Summing $f_{\mathcal{V}}(\hat{x}') - f_{\mathcal{V}}(\hat{x}) = \sum_{j=1}^{r}(t_j - t_{j-1})\big[\alpha h_{r_k}'(t_j^\star) v_{r_k} + \alpha \sum_{s\in\mathcal{F}_k} g_s'(t_j^\star) \sum_{b\in B(s)} v_b\big]$ and using $\|\hat{x}' - \hat{x}\|_2 \leq \|x' - x\|_2$ gives

$$\big\|f_{\mathcal{V}}(x) - f_{\mathcal{V}}(x')\big\|_2 \;\leq\; \left(\frac{\alpha}{\delta_w} + \alpha(k+1)\big(10^{10}k^6\big)\frac{2}{\delta_w}\right)\|x - x'\|_2 \;<\; \big(8 \cdot 10^{11}k^8\big)\|x - x'\|_2.$$

Using $|\frac{1}{4}\|u\|_2^2 - \frac{1}{4}\|v\|_2^2| \leq \|u - v\|_2$ for $u, v \in B_{d_0}(\mathbf{0}, 2)$, both c_1 and c_2 are Euclidean L-Lipschitz with $L := 10^{12}k^8 = d_0^{1/2} < d^{1/2}$.

(3) Testing $\|z\|_2^2 \leq 1$ takes $2d_i$ scalar operations; if $\|z\|_2^2 > 1$, computing $m := \max_{1\leq j\leq d_i} \max\{z_j, 0 - z_j\} = \|z\|_\infty > 0$ and returning $h(z) := z/m$ ($\|h(z)\|_\infty = 1$, $\max_{y\in B_{d_i}(\mathbf{0},1)} \langle h(z), y\rangle \leq \frac{\|z\|_2}{\|z\|_\infty} < \frac{\|z\|_2^2}{\|z\|_\infty} = \langle h(z), z\rangle$) uses $4d_i - 1$ additional scalar operations ($6d_i - 1$ total). □

Before analyzing arbitrary correlated distributions, we verify that pure iteration from $x^{(0)} = \mathbf{0}$ traces all $N = 3 \cdot 2^k - 2$ vertices of $\mathcal{F}_k$ in post-order before terminating at the unique pure Nash equilibrium $(\alpha v_{r_k}, \alpha v_{r_k})$.

Lemma 6 (Exact N-Step Post-Order Trajectory and Terminal Pure Nash Equilibrium)**.** *For any μ_{coh}-incoherent tuple $\mathcal{V}$, starting from $x^{(0)} := \mathbf{0} \in \mathbb{R}^{d_0}$, the iterates $x^{(t)} := f_{\mathcal{V}}(x^{(t-1)})$ satisfy:*

1. *For every $t \in \{0, \dots, N-1\}$, $x^{(t)} = \alpha \sum_{b \in \mathrm{Req}(s_{t+1})} v_b \in B_{d_0}(\mathbf{0}, \frac{1}{16})$, with $L_{s_{t+1}}(w(x^{(t)})) = 1$ and $L_s(w(x^{(t)})) = 0$ for all $s \neq s_{t+1}$;*
2. *At step N, $x^{(N)} = \alpha v_{r_k}$ satisfies $f_{\mathcal{V}}(x^{(N)}) = x^{(N)}$, so $((x^{(N)}, \mathbf{0}), (x^{(N)}, \mathbf{0}))$ is an exact pure Nash equilibrium of $G_{\mathcal{V},d}$ with zero cost for both players.*

Proof. Set $B_t := \mathrm{Req}(s_{t+1})$ $(0 \le t \le N-1)$ and $B_N := \{r_k\}$. By Lemma 1(1), $B_0 = \emptyset$, $B(s_{t+1}) = B_{t+1}$ $(t \le N-2)$, $B(s_N) = B_N$, and $|B_t| \le k+1$, so $\|x^{(t)}\|_2 \le \alpha(k+1) = \frac{1}{16}$ at every step. Since $\alpha - k\alpha\mu_{\mathrm{coh}} > \frac{3}{4}\alpha > \tau_0 + \delta_w$ and $(k+1)\alpha\mu_{\mathrm{coh}} < \tau_0$, at step t we have $w_a(x^{(t)}) = \mathbb{I}[a \in B_t]$ for all $a \in \mathcal{F}_k$. When $t < N$, disjointness $\mathrm{Req}(s_{t+1}) \cap \mathrm{Block}(s_{t+1}) = \emptyset$ and $r_k \in \mathrm{Block}(s_{t+1})$ yield $w_{r_k}(x^{(t)}) = 0$ and $L_{s_{t+1}}(w(x^{(t)})) = 1$, so Lemma 1(2) forces $L_s(w(x^{(t)})) = 0$ for all $s \neq s_{t+1}$ and $x^{(t+1)} = \alpha \sum_{b \in B_{t+1}} v_b$. At $t = N$, $w_{r_k}(x^{(N)}) = 1$ gives $f_{\mathcal{V}}(x^{(N)}) = \alpha v_{r_k} = x^{(N)}$. □

5 Hilbert-Space Adjoint Contraction and Non-Conservative Drift

We now show that any correlated distribution $\mu \in \Delta_{\mathrm{fin}}(\mathcal{X}_1 \times \mathcal{X}_2)$ with swap gap at most ϵ projects to a distribution μ_{proj} on active coordinates whose two-step conditional expectation operator $A = PP^\star$ satisfies $\|X_{\mathrm{act}} - Af_{\mathcal{V}}(X_{\mathrm{act}})\|_{L^2(\mu_{1,\mathrm{proj}})} \le 4\sqrt{\epsilon}$.

Lemma 7 (Active Projection Monotonicity and Hilbert-Space Adjoint Contraction)**.** *Let $\mu \in \Delta_{\mathrm{fin}}(\mathcal{X}_1 \times \mathcal{X}_2)$ have $\epsilon := \max\{\mathrm{SwapGap}_1(\mu), \mathrm{SwapGap}_2(\mu)\}$, and let $\mu_{\mathrm{proj}} \in \Delta_{\mathrm{fin}}\big(B_{d_0}(\mathbf{0}, 1)^2\big)$ denote the active-coordinate pushforward of $(X_{\mathrm{act}}, Y_{\mathrm{act}})$ under $(X, Y) \sim \mu$, with marginals $\mu_{1,\mathrm{proj}}, \mu_{2,\mathrm{proj}}$.*

1. ***(Swap-gap monotonicity):** In $G_{\mathcal{V}} = (B_{d_0}(\mathbf{0}, 1), B_{d_0}(\mathbf{0}, 1), c_1^{\mathrm{act}}, c_2^{\mathrm{act}})$, each player $i \in \{1, 2\}$ satisfies $\mathrm{SwapGap}_i(\mu_{\mathrm{proj}}) \le \mathrm{SwapGap}_i(\mu) \le \epsilon$.*
2. ***(Two-step L^2 residual bound):** Define total maps $P : L^2(\mu_{2,\mathrm{proj}}; \mathbb{R}^{d_0}) \to L^2(\mu_{1,\mathrm{proj}}; \mathbb{R}^{d_0})$ and $P^\star : L^2(\mu_{1,\mathrm{proj}}; \mathbb{R}^{d_0}) \to L^2(\mu_{2,\mathrm{proj}}; \mathbb{R}^{d_0})$ by $(Ph)(u) := \mathbb{E}_{\mu_{\mathrm{proj}}}[h(Y_{\mathrm{act}}) \mid X_{\mathrm{act}} = u]$ for $u \in \mathrm{supp}(\mu_{1,\mathrm{proj}})$ (and $\mathbf{0}$ otherwise) and $(P^\star g)(v) := \mathbb{E}_{\mu_{\mathrm{proj}}}[g(X_{\mathrm{act}}) \mid Y_{\mathrm{act}} = v]$ for $v \in \mathrm{supp}(\mu_{2,\mathrm{proj}})$ (and $\mathbf{0}$ otherwise). Setting $A := PP^\star$,*

$$\big\|X_{\mathrm{act}} - Af_{\mathcal{V}}(X_{\mathrm{act}})\big\|_{L^2(\mu_{1,\mathrm{proj}}; \mathbb{R}^{d_0})} \;\le\; 4\sqrt{\epsilon}.$$

3. ***(Self-adjoint two-step symmetrization):** Drawing $(X_{\mathrm{act}}, Y_{\mathrm{act}}) \sim \mu_{\mathrm{proj}}$ and $X'_{\mathrm{act}} \sim \mu_{\mathrm{proj}}(\cdot \mid Y_{\mathrm{act}})$ conditionally independently given Y_{act}, for any $F, H : B_{d_0}(\mathbf{0}, 1) \to \mathbb{R}^{d_0}$,*

$$\begin{aligned}&\Big\langle F(X_{\mathrm{act}}),\ A\,H(X_{\mathrm{act}})\Big\rangle_{L^2(\mu_{1,\mathrm{proj}})}\\ &\quad= \frac{1}{2}\,\mathbb{E}_{X_{\mathrm{act}} \leftarrow Y_{\mathrm{act}} \rightarrow X'_{\mathrm{act}}}\Big[\big\langle F(X_{\mathrm{act}}), H(X'_{\mathrm{act}})\big\rangle + \big\langle F(X'_{\mathrm{act}}), H(X_{\mathrm{act}})\big\rangle\Big].\end{aligned}$$

Proof. **(1)** Any active deviation $\phi_i^{\mathrm{act}} : B_{d_0}(\mathbf{0}, 1) \to B_{d_0}(\mathbf{0}, 1)$ lifts to a unit-ball-feasible deviation $\phi_i(z_{\mathrm{act}}, z_{\mathrm{pad}}) := (\phi_i^{\mathrm{act}}(z_{\mathrm{act}}), \mathbf{0}) \in B_{d_i}(\mathbf{0}, 1)$ on $\mathcal{X}_i$. Since $\frac{1}{4}\|X_{\mathrm{pad}}\|_2^2 \ge 0$ and $\frac{1}{4}\|Y_{\mathrm{pad}}\|_2^2 \ge 0$, zeroing out the padding coordinates only decreases player i's cost further, so $\mathrm{SwapGap}_i(\mu_{\mathrm{proj}}) \le \mathrm{SwapGap}_i(\mu) \le \epsilon$.

(2) Write $(U, V) := (X_{\mathrm{act}}, Y_{\mathrm{act}})$. By convexity of $B_{d_0}(\mathbf{0}, 1)$ and Lemma 5(1), both $\phi_1^{\mathrm{act}}(u) := (PV)(u)$ and $\phi_2^{\mathrm{act}}(v) := (P^\star f_{\mathcal{V}}(U))(v)$ map $B_{d_0}(\mathbf{0}, 1)$ into $B_{d_0}(\mathbf{0}, 1)$. By conditional orthogonality

$\mathbb{E}[\langle U - PV(U), PV(U) - V\rangle] = 0$ and $\mathbb{E}[\langle V - P^\star f_{\mathcal{V}}(U)(V), P^\star f_{\mathcal{V}}(U)(V) - f_{\mathcal{V}}(U)\rangle] = 0$, testing ϕ_1^{act} and ϕ_2^{act} gives $\frac{1}{4}\|U - PV\|^2_{L^2(\mu_{1,\mathrm{proj}})} \le \epsilon$ and $\frac{1}{4}\|V - P^\star f_{\mathcal{V}}(U)\|^2_{L^2(\mu_{2,\mathrm{proj}})} \le \epsilon$. Since $\|P\|_{L^2\to L^2} \le 1$ by conditional Jensen's inequality, the triangle inequality gives $\|U - PP^\star f_{\mathcal{V}}(U)\|_{L^2(\mu_{1,\mathrm{proj}})} \le \|U - PV\|_{L^2} + \|P(V - P^\star f_{\mathcal{V}}(U))\|_{L^2} \le 4\sqrt{\epsilon}$.

(3) Conditional independence of U and $U' := X'_{\mathrm{act}}$ given V makes $(U, U') \overset{d}{=} (U', U)$ symmetric with $\mathbb{E}[\langle F(U), H(U')\rangle] = \mathbb{E}_V[\langle (P^\star F)(V), (P^\star H)(V)\rangle] = \langle F(U), PP^\star H(U)\rangle_{L^2(\mu_{1,\mathrm{proj}})}$. □

We now contract $Af_{\mathcal{V}}(X_{\mathrm{act}}) - X_{\mathrm{act}}$ against the non-conservative test vector field $G(x) = \sum_{s\in\mathcal{F}_k} L_s(w(x))u_s$, showing that any joint profile placing negligible mass on $\{|\langle x_{\mathrm{act}}, v_{r_k}\rangle| > \tau_0\}$ must suffer swap gap strictly greater than $\frac{1}{d}$.

Lemma 8 (Non-Conservative Drift and Swap-Gap Lower Bound)**.** *Let $d \ge d_{\min}$ ($k = k(d) \ge 2$), let $\mathcal{V}$ be μ_{coh}-incoherent, and let $\mu \in \Delta_{\mathrm{fin}}(\mathcal{X}_1 \times \mathcal{X}_2)$ satisfy*

$$\mu_{1,\mathrm{proj}}\Big(\big\{x \in B_{d_0}(\mathbf{0}, 1) : |\langle x, v_{r_k}\rangle| > \tau_0\big\}\Big) \;\le\; 2^{-10k}.$$

Then

$$\max\big\{\mathrm{SwapGap}_1(\mu),\ \mathrm{SwapGap}_2(\mu)\big\} \;\ge\; \frac{\alpha^2}{512(2k+1)} \;>\; \frac{1}{d}.$$

Proof. Let $\epsilon := \max\{\mathrm{SwapGap}_1(\mu), \mathrm{SwapGap}_2(\mu)\}$, write $(U, V) := (X_{\mathrm{act}}, Y_{\mathrm{act}}) \sim \mu_{\mathrm{proj}}$, and define for each $s \in \mathcal{F}_k$ the path sum $u_s := \sum_{a\in\mathrm{Path}(s,r_k)} v_a$ and the test field $G : B_{d_0}(\mathbf{0}, 1) \to \mathbb{R}^{d_0}$ by

$$G(x) \;:=\; \sum_{s\in\mathcal{F}_k} L_s\big(w(x)\big)\, u_s \;=\; \sum_{a\in\mathcal{F}_k} S_a\big(w(x)\big)\, v_a.$$

Since $|\mathrm{Path}(s, r_k)| \le 2k{+}1$ and $2k\,\mu_{\mathrm{coh}} < 1$, every $s \in \mathcal{F}_k$ satisfies $\|u_s\|_2^2 \le |\mathrm{Path}(s, r_k)|(1{+}2k\,\mu_{\mathrm{coh}}) < 2(2k + 1)$, so Lemma 1(2) gives $\|G(x)\|_2 \le \sqrt{2(2k+1)}$ everywhere on $B_{d_0}(\mathbf{0}, 1)$. Applying Cauchy–Schwarz and Lemma 7(2)–(3),

$$\begin{aligned} 4\sqrt{2(2k+1)}\,\sqrt{\epsilon} \;&\ge\; \Big\langle G(U),\ Af_{\mathcal{V}}(U) - U\Big\rangle_{L^2(\mu_{1,\mathrm{proj}})} \\ &=\; \frac{1}{2}\,\mathbb{E}_{U\leftarrow V\to U'}\Big[\big\langle G(U), f_{\mathcal{V}}(U')\big\rangle + \big\langle G(U'), f_{\mathcal{V}}(U)\big\rangle\Big] \;-\; \mathbb{E}_{\mu_{1,\mathrm{proj}}}\Big[\big\langle G(U), U\big\rangle\Big]. \end{aligned}$$

Let $E_0 := \{x \in B_{d_0}(\mathbf{0}, 1) : |\langle x, v_{r_k}\rangle| \le \tau_0\}$ denote the pre-terminal slab where the root coordinate v_{r_k} has not yet activated ($w_{r_k}(x) = 0$). We bound both terms deterministically across E_0:

1. **Pointwise upper bound on $\langle G(x), x\rangle$ for $x \in E_0$:** For $a = r_k$, $S_{r_k}(w(x)) \le 1$ gives $S_{r_k}(w(x))\langle x, v_{r_k}\rangle \le \tau_0$. For any $a \in \mathcal{F}_k \setminus \{r_k\}$, whenever $S_a(w(x)) > 0$, ancestor complementarity (Lemma 1(3)) gives $1 - w_a(x) \ge S_a(w(x)) > 0$, so $w_a(x) < 1$ and therefore $\langle x, v_a\rangle < \tau_0 + \delta_w$. Since $\sum_{a\in\mathcal{F}_k} S_a(w(x)) \le 2k + 1$, every $x \in E_0$ satisfies

$$\begin{aligned} \big\langle G(x), x\big\rangle \;&=\; \sum_{a\in\mathcal{F}_k} S_a(w(x))\langle x, v_a\rangle \;\le\; \tau_0 + (2k+1)(\tau_0 + \delta_w) \\ &=\; \left(\frac{1}{128(2k+1)} + \frac{17}{128}\right)\alpha \;<\; \frac{3}{16}\,\alpha. \end{aligned}$$

2. **Pointwise lower bound on $\frac{1}{2}\big(\langle G(x), f_{\mathcal{V}}(x')\rangle + \langle G(x'), f_{\mathcal{V}}(x)\rangle\big)$ for $(x, x') \in E_0 \times E_0$:** For any $x, x' \in E_0$, $w_{r_k}(x) = w_{r_k}(x') = 0$ and $\sum_{s\in\mathcal{F}_k} L_s(w(x)) = \sum_{s'\in\mathcal{F}_k} L_{s'}(w(x')) = 1$. By Lemma 3, $|\mathrm{Path}(s, r_k)\cap B(s')| = \mathbb{I}[s \le_{\mathrm{post}} s']$, and off-diagonal inner products across at most $(2k+1)(k+1)$ pairs have magnitude at most $(2k+1)(k+1)\mu_{\mathrm{coh}} < \frac{1}{64}$. Since $\mathbb{I}[s \le_{\mathrm{post}} s'] + \mathbb{I}[s' \le_{\mathrm{post}} s] \ge 1$ holds for every pair $(s, s') \in \mathcal{F}_k^2$, symmetrizing and summing over $(s, s') \in \mathcal{F}_k^2$ yields

$$\frac{1}{2}\Big(\big\langle G(x), f_{\mathcal{V}}(x')\big\rangle + \big\langle G(x'), f_{\mathcal{V}}(x)\big\rangle\Big) \;\ge\; \frac{\alpha}{2} - \alpha(2k+1)(k+1)\mu_{\mathrm{coh}} \;>\; \frac{31}{64}\,\alpha$$

(giving a pointwise symmetrized drift $\frac{1}{2}(\langle G(x), f_{\mathcal{V}}(x') - x\rangle + \langle G(x'), f_{\mathcal{V}}(x) - x'\rangle) > \frac{19}{64}\alpha > \frac{9}{32}\alpha$ across all $(x, x') \in E_0 \times E_0$).

Outside $E_0 \times E_0$ (which occurs with probability at most $2\,\mu_{1,\mathrm{proj}}(E_0^c) \le 2 \cdot 2^{-10k}$), Cauchy–Schwarz and $\|f_{\mathcal{V}}\|_2 \le \frac{1}{16}$ bound $|\langle G(x), x\rangle| \le \sqrt{2(2k+1)}$ and $\frac{1}{2}|\langle G(x), f_{\mathcal{V}}(x')\rangle + \langle G(x'), f_{\mathcal{V}}(x)\rangle| \le \frac{1}{16}\sqrt{2(2k+1)}$. Hence

$$4\sqrt{2(2k+1)}\,\sqrt{\epsilon} \;\ge\; \frac{31}{64}\,\alpha\big(1 - 2\cdot 2^{-10k}\big) - \frac{3}{16}\,\alpha - \frac{9}{8}\cdot 2^{-10k}\sqrt{2(2k+1)} \;>\; \frac{\alpha}{4},$$

so $\epsilon \ge \frac{\alpha^2}{512(2k+1)} = \frac{1}{131\,072(k+1)^2(2k+1)} > \frac{1}{3\cdot 10^{24}(k+1)^{16}} > \frac{1}{d}$. □

6 Oracle Simulation and Proof of Main Theorem

Finally, we show that each query in model **(M1)** reveals at most one additional post-order vector v_{s_q} along $\mathcal{F}_k$. By enforcing a $\frac{\tau_0}{2}$ inner-product margin against all unrevealed directions $\{v_{s_q}, \dots, v_{s_N}\}$, the true map $f_{\mathcal{V}}$ and the q-step prefix truncation agree identically on an entire open ball of radius $\frac{\tau_0}{2}$ around the q-th query point—so even an oracle returning arbitrary higher-order local derivatives with respect to both players' coordinates learns zero information about $v_{r_k} = v_{s_N}$ during the first $Q \le N - 1$ queries.

Lemma 9 (Prefix-Only Oracle Simulation)**.** *Let $d \ge d_{\min}$ ($k = k(d) \ge 2$) and draw $\mathcal{V} \sim \pi := \mathrm{Unif}\big(\{\pm d_0^{-1/2}\}^{d_0\times\mathcal{F}_k}\big)$. For any deterministic algorithm $\mathcal{A}$ in model **(M1)** making at most $Q \le N-1 = 3\cdot 2^k - 3$ first-order queries $(x_{\mathrm{q}}^{(q)}, y_{\mathrm{q}}^{(q)})_{q=1}^{Q}$ (plus any separation queries $\mathrm{SEP}_{\mathcal{X}_i}$) and outputting $\mu = \sum_{j=1}^{M} p_j \delta_{(x^{(j)}, y^{(j)})} \in \Delta_{\mathrm{fin}}(\mathcal{X}_1 \times \mathcal{X}_2)$, with probability at least $1 - 2^{-4k}$:*

1. *$\mathcal{V}$ is μ_{coh}-incoherent ($|\langle v_a, v_b\rangle| \le \mu_{\mathrm{coh}}$ for all $a \ne b \in \mathcal{F}_k$);*
2. *At every query $q \in \{1, \dots, Q\}$, $\big|\langle \Pi_{B_{d_0}(\mathbf{0},1)}(x_{\mathrm{q,act}}^{(q)}), v_a\rangle\big| \le \frac{\tau_0}{2}$ holds for every $a \in \{s_q, \dots, s_N\}$, so $f_{\mathcal{V}}(z) = \alpha \sum_{s\in\{s_1,\dots,s_q\}} L_s(w^{(q)}(\hat{z})) \sum_{b\in B(s)} v_b$ holds on the open ball $B_{d_0}\big(\Pi_{B_{d_0}(\mathbf{0},1)}(x_{\mathrm{q,act}}^{(q)}), \frac{\tau_0}{2}\big)$;*
3. *The output active marginal satisfies $\mu_{1,\mathrm{proj}}\big(\{x \in B_{d_0}(\mathbf{0}, 1) : |\langle x, v_{r_k}\rangle| > \tau_0\}\big) \le 2^{-10k}$.*

Proof. Define a prefix-simulated run of $\mathcal{A}$ inductively across steps $q = 1, \dots, Q$:

- Every separation query $\mathrm{SEP}_{\mathcal{X}_i}(z)$ depends only on z and d_i (Lemma 5(3)), independent of $\mathcal{V}$.
- Before step q ($1 \le q \le Q$), the simulated answers to queries $1, \dots, q-1$ depend solely on $(v_{s_1}, \dots, v_{s_{q-1}})$ (the empty tuple for $q = 1$). Consequently, the q-th simulated query $(\tilde{x}_{\mathrm{q}}^{(q)}, \tilde{y}_{\mathrm{q}}^{(q)})$—and hence its clamped active component $\hat{u}_{\mathrm{sim}}^{(q)} := \Pi_{B_{d_0}(\mathbf{0},1)}(\tilde{x}_{\mathrm{q,act}}^{(q)}) \in B_{d_0}(\mathbf{0}, 1)$—is a deterministic function of $(v_{s_1}, \dots, v_{s_{q-1}})$ alone, independent of $(v_{s_q}, \dots, v_{s_N})$.

- To answer query q, set $w_a^{(q)}(\hat{z}) := w_a(\hat{z})$ for $a \in \{s_1, \dots, s_q\}$ and $w_a^{(q)}(\hat{z}) := 0$ for $a \in \{s_{q+1}, \dots, s_N\}$, and evaluate costs and gradients using the truncated map

$$f_{\mathcal{V}}^{(q)}(z) \;:=\; \alpha \sum_{s \in \{s_1,\dots,s_q\}} L_s\big(w^{(q)}(\hat{z})\big) \sum_{b \in B(s)} v_b.$$

  Since $B(s) \subseteq \{s_1, \dots, s_q\}$ for every $s \in \{s_1, \dots, s_q\}$ by Lemma 1(1), the q-th simulated answer depends solely on $(v_{s_1}, \dots, v_{s_q})$, maintaining the induction invariant.

Let $E_{\mathrm{sim}} := \bigcap_{q=1}^{Q} \bigcap_{a \in \{s_q,\dots,s_N\}} \big\{ |\langle \hat{u}_{\mathrm{sim}}^{(q)}, v_a \rangle| \le \frac{\tau_0}{2} \big\}$. By Lemma 4(1) conditional on $(v_{s_1}, \dots, v_{s_{q-1}})$ and a union bound over at most $QN < N^2 < 9 \cdot 2^{2k}$ pairs (q, a), $\Pr[\neg E_{\mathrm{sim}}] < 9 \cdot 2^{2k} \cdot 2^{-30k}$. On E_{sim}, for every z with $\|\hat{z} - \hat{u}_{\mathrm{sim}}^{(q)}\|_2 < \frac{\tau_0}{2}$ and every $a \in \{s_q, \dots, s_N\}$ (which includes $r_k = s_N$), $|\langle \hat{z}, v_a \rangle| < \tau_0$, so $w_a(\hat{z}) = 0 = w_a^{(q)}(\hat{z})$; moreover, by prefix blocking (Lemma 2(2)), every $b \in \{s_{q+1}, \dots, s_N\}$ has an element in $\mathrm{Req}(b) \cap \{s_q, \dots, s_N\}$ with zero ramp weight, forcing $L_b(w(\hat{z})) = 0$. Hence $f_{\mathcal{V}}(z) = f_{\mathcal{V}}^{(q)}(z)$ on an open neighborhood of $\tilde{x}_{\mathrm{q,act}}^{(q)}$, ensuring that the true and simulated runs produce identical queries and identical output distributions $\mu = \mu^{\mathrm{sim}} = \sum_{j=1}^{M} p_j^{\mathrm{sim}} \delta_{(x_{\mathrm{sim}}^{(j)}, y_{\mathrm{sim}}^{(j)})}$ across all steps.

Finally, because $Q \le N - 1$, the simulated output μ^{sim} depends only on $(v_{s_1}, \dots, v_{s_{N-1}})$ and is independent of $v_{r_k} = v_{s_N}$. Conditioning on $(v_{s_1}, \dots, v_{s_{N-1}})$, Lemma 4(1) gives for each atom $x_{\mathrm{sim,act}}^{(j)} \in B_{d_0}(\mathbf{0}, 1)$,

$$\mathbb{E}_{v_{r_k}}\Big[\mu_{1,\mathrm{proj}}^{\mathrm{sim}}\big(\{|\langle x, v_{r_k}\rangle| > \tau_0\}\big)\Big] \;=\; \sum_{j=1}^{M} p_j^{\mathrm{sim}} \Pr_{v_{r_k}}\Big[\big|\langle x_{\mathrm{sim,act}}^{(j)}, v_{r_k}\rangle\big| > \tau_0\Big] \;<\; 2^{-30k}.$$

By Markov's inequality, $\Pr\big[\mu_{1,\mathrm{proj}}^{\mathrm{sim}}(\{|\langle x, v_{r_k}\rangle| > \tau_0\}) > 2^{-10k}\big] < 2^{-20k}$. Union-bounding with E_{sim} and Lemma 4(2) bounds the total failure probability by $2^{-8k} + 9 \cdot 2^{2k} \cdot 2^{-30k} + 2^{-20k} < 2^{-4k}$. □

Theorem 2 (Unconditional Superpolynomial Black-Box Oracle Lower Bound)**.** *In the black-box oracle model **(M1)**, computing an ϵ-approximate normal-form correlated equilibrium of two-player convex games at inverse-linear precision $\epsilon = \frac{1}{d}$ admits no FPTAS. Specifically, for every integer $d \ge d_{\min} := 2 \cdot 10^{24} \cdot 2^{16}$ with $d_1 = \lfloor d/2 \rfloor$, $d_2 = \lceil d/2 \rceil$, and $k = k(d) := \Big\lfloor \big(\frac{d_1}{10^{24}}\big)^{1/16} \Big\rfloor \ge 2$:*

1. *Every instance $G_{\mathcal{V},d} = (\mathcal{X}_1, \mathcal{X}_2, c_1, c_2)$ with μ_{coh}-incoherent unit vectors $\mathcal{V}$ is a valid two-player convex game on exact unit balls $\mathcal{X}_i = B_{d_i}(\mathbf{0}, 1)$ ($R = 1$), with cost range $[0, 1]$, strict $\frac{1}{2}$-strong convexity in own strategy ($\nabla_{x_i}^2 c_i = \frac{1}{2} I_{d_i}$), Euclidean Lipschitz constant $L = 10^{12} k^8 = d_0^{1/2} < d^{1/2}$, and deterministic strong separation oracle $\mathrm{SEP}_{\mathcal{X}_i}$ independent of $\mathcal{V}$.*

2. *At precision $\epsilon := \frac{1}{d}$, for every randomized algorithm $\mathcal{R}$ making at most*

$$Q_0 \;\le\; 3 \cdot 2^{k(d)} - 3 \qquad \left(\text{and in particular } Q_0 \;\le\; 2^{k(d)-1} \;>\; \frac{1}{4} \exp\!\left(\frac{\ln 2}{(3 \cdot 10^{24})^{1/16}}\, d^{1/16} \right) \right)$$

   first-order game queries, regardless of internal computation time or output support size $M = |\operatorname{supp}(\mu)| \in \mathbb{N}$, there exists a deterministic valid instance $G_{\mathcal{V}^\star,d}$ on which $\mathcal{R}$ fails to output an ϵ-CE with probability at least $\frac{7}{8} > \frac{1}{3}$.

Proof. Let $\mathcal{I}_{k,d} \subseteq \{\pm d_0^{-1/2}\}^{d_0 \times \mathcal{F}_k}$ be the nonempty family of μ_{coh}-incoherent Rademacher tuples ($\Pr_\pi[\mathcal{V} \in \mathcal{I}_{k,d}] \ge 1 - 2^{-8k}$ by Lemma 4(2)), each satisfying (1) by Lemma 5. For any randomized

algorithm $\mathcal{R}$ making at most $Q_0 \le 3 \cdot 2^k - 3 = N - 1$ queries, Yao's Minimax Principle (External Result 1) on $\mathrm{Unif}(\mathcal{I}_{k,d})$ combined with Lemma 9 and Lemma 8 bounds the worst-case success probability of $\mathcal{R}$ by

$$\inf_{\mathcal{V} \in \mathcal{I}_{k,d}} \Pr_{\mathcal{R}} \Big[\mu \text{ is an } \epsilon\text{-CE of } G_{\mathcal{V},d} \Big] \;\le\; \frac{2^{-4k}}{1 - 2^{-8k}} \;<\; \frac{1}{8},$$

so $\mathcal{R}$ fails with probability at least $\frac{7}{8}$ on at least one deterministic instance $G_{\mathcal{V}^\star,d}$ ($\mathcal{V}^\star \in \mathcal{I}_{k,d}$). $\square$

6.1 Reconciling Our Lower Bound with Known Positive Results

1. **Constant-ϵ PTAS via `TreeSwap` [7, 3]:** For any fixed constant $\epsilon \in (0,1)$, running no-swap-regret dynamics via `TreeSwap` from $(\mathbf{0}, \mathbf{0})$ uses $\exp(\tilde{O}(1/\epsilon)) = O_\epsilon(1)$ oracle calls. In our construction, Lemma 8 yields swap gap $\Omega(\alpha^2/k) = \Theta(k^{-3})$ prior to reaching r_k; whenever d grows past $\Theta(\epsilon^{-16/3})$, our step scale $\alpha = \frac{1}{16(k+1)}$ drops below $\epsilon^{1/3}$, allowing constant-precision ϵ-CE profiles supported on early vertices of $\mathcal{F}_k$. At inverse-linear precision $\epsilon = \frac{1}{d}$, by contrast, $\frac{\alpha^2}{512(2k+1)} > \frac{1}{d}$ holds for *every* $d \ge d_{\min}$, forcing every $\frac{1}{d}$-CE to discover v_{r_k} after $> 3 \cdot 2^{k(d)} - 3$ queries.

2. **Polynomial-Time Algorithms for Coarse CE, Linear CE, and Φ-Equilibria [5, 4, 1]:** Along the M-step pure trajectory $\bar{\mu}_M := \frac{1}{M} \sum_{t=0}^{M-1} \delta_{((x^{(t)},\mathbf{0}),(x^{(t+1)},\mathbf{0}))}$, every smooth scalar potential $\Phi : B_{d_0}(\mathbf{0}, 1) \to [-1, 1]$ telescopes across steps as $\frac{1}{M} \sum_{t=0}^{M-1} \big(\Phi(x^{(t+1)}) - \Phi(x^{(t)})\big) = \frac{\Phi(x^{(M)}) - \Phi(x^{(0)})}{M} \le \frac{2}{M}$, vanishing in $M = O(1/\epsilon)$ steps. At the same time, because consecutive iterates differ by step length $\Theta(\alpha) = \Theta(d^{-1/16})$, the full normal-form swap gap of $\bar{\mu}_M$ remains $\mathrm{SwapGap}_1(\bar{\mu}_M) = \frac{1}{4M} \sum_{t=0}^{M-1} \|x^{(t)} - x^{(t+1)}\|_2^2 = \Theta(\alpha^2) \gg \frac{1}{d}$ for all $M < N$. While low-dimensional Φ-regret minimizers and Ellipsoid-Against-Hope algorithms find coarse, linear, and polynomial-dimensional Φ-equilibria of $G_{\mathcal{V},d}$ in $\mathrm{poly}(d, 1/\epsilon)$ queries [4, 1], no polynomial-query algorithm can suppress regret against the full N-state non-conservative deviation field $G(x) = \sum_{s \in \mathcal{F}_k} L_s(w(x)) u_s$.

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